



CD/JUL 2024/ISP550/ITS572

**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	INFORMATION SYSTEMS ENGINEERING
COURSE CODE	:	ISP550/ITS572
EXAMINATION	:	JULY 2024
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of 2 parts: PART A (10 Questions)
PART B (4 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A

State whether each of the following statement is **TRUE** or **FALSE**.

1. System design requires careful study of current systems, manual and computerized, that might be replaced or enhanced.
2. IT services firms only help companies develop custom information systems for internal use.
3. In System Development Life Cycle (SDLC), adaptive approach is more flexible as compared to predictive approach.
4. Documentation is one of the criteria to consider when choosing off-the-shelf software.
5. An association is a relationship between things in the problem domain.
6. CRUD is a technique to validate, refine or cross-check use cases.
7. Analysis says "how" is required and design tells us "what" the system will be configured and constructed.
8. Reliability and physical facilities are the only issues when considering hosting alternatives.
9. System documentation is a description of how to interact with and maintain the system.
10. Without conducting a user training session, user error rates will be high.

(10 marks)

PART B**QUESTION 1**

- a) Business and systems have become more sophisticated and more complex, and there has been an increasing emphasis on speed in systems analysis and design.

Describe **THREE (3)** factors that affect the selection of system development methodology.

(6 marks)

- b) Describe **TWO (2)** reasons why a company would need to have an awareness of outsourcing as an alternative to handling IT services in-house.

(6 marks)

- c) In-house development allows organizations to design systems that align perfectly with their business processes. Explain when an organization should choose in-house developers to develop information systems.

(4 marks)

QUESTION 2

- a) List the **TWO (2)** common skills and **TWO (2)** common activities of a project manager.

(4 marks)

- b) When developing a communication plan, many questions must be answered to ensure that the plan is comprehensive and complete. List the **SIX (6)** required questions.

(6 marks)

- c) Effective project identification and selection is essential if organizations are to achieve the greatest benefits from information systems.

- i) Describe **THREE (3)** intangible benefits from the information system development.

(6 marks)

- ii) Describe **TWO (2)** possible evaluation criteria when classifying and ranking projects.

(4 marks)

- d) The purpose of system testing is to confirm that the system satisfies requirements. Mosley (1993) organizes the seven types of tests according to whether they employ static or dynamic techniques and whether the test is automated or manual.

Describe Unit Testing, Desk Checking and System Testing.

(6 marks)

QUESTION 3

Unified Modeling Language (UML) activity diagrams and sequence diagrams are two examples of requirements model.

- a) Explain the difference of purpose between UML activity diagram and sequence diagram. (4 marks)

- b) Develop an activity diagram based on the following narrative.

The purpose of the Open Access Insurance System is to provide automotive insurance to car owners. Initially, prospective customers fill out an insurance application, which provides information about the customer and his or her vehicles. This information is sent to an agent, who sends it to various insurance companies to get quotes for insurance. When the responses return, the agent then determines the best policy for the type and level of coverage desired and gives the customer a copy of the insurance policy proposal and quote.

(8 marks)

- c) Designing the interfaces (inputs and outputs) between a system and its users and environment is an important system development task.

- i) Give **TWO (2)** results of the interfaces that are poorly designed.

(3 marks)

- ii) A good user interface design anticipates common errors and helps the user to avoid them. When the system finds an error, the error message should state specifically what is wrong and explain how to correct it. Give **TWO (2)** examples of good error message.

(3 marks)

- d) Controls are mechanisms and procedures that are built into a system to safeguard the system and the information within it. Describe **THREE (3)** primary objectives of integrity controls.

(6 marks)

QUESTION 4

- a) One of the toughest problems in investigating system requirements is to ensure that they are complete and comprehensive. Describe **TWO (2)** ways an analyst would do to get all the right information during an interview session.

(4 marks)

- b) Describe **TWO (2)** most appropriate information-gathering methods to understand the business requirements for a Pharmaceutical Company's management staff, client, and suppliers.

(6 marks)

- c) Al-Farabi Dental Clinic needs a system that keeps patients records. The clinic's staff has provided information about the process in the form of notes. Your role as the analyst is to analyze the notes.

The receptionist keeps track of patients and head of household information. He/she will enter information about the patients and head of household. He/she will also keep track of clinic visits by the patients. Patient information is also entered and maintained by the clinic business manager. In addition, the business manager maintains the information about the dental staff persons. The business manager also prints the invoices. Patient invoices are printed monthly and emailed to the head of the household. Insurance invoices are printed weekly. When the invoices are printed, the business manager double-checks a few invoices against information in the system to make sure it is being aggregated correctly. The business manager also enters the payment information when it is received. Each dental staff person is responsible for entering information about the dental procedures that he/she performs. The business manager also prints an overdue invoice report showing heads of household who are behind on their payments. Sometimes dentists like to see a list of the procedures they performed during a week or month, and they can view that report.

Construct a use case diagram for the above descriptions.

(8 marks)

- d) Assume that you are in charge of implementation and deployment of a new system that is replacing a critical existing system that is used 24 hours a day. To minimize risk, you plan to phase the deployment of new subsystems over a period of six weeks and operate both systems in parallel for at least three weeks beyond the last new subsystem deployment. Because there aren't enough personnel to operate both systems, you plan to hire up to 30 temporary workers during the parallel operation period.

Describe **THREE (3)** tasks that you should assign to the trained temporary workers.

(6 marks)

END OF QUESTION PAPER